ETL SCD1 using Snowflake Tasks, Streams and Stored Procedures

Objective

This project implements a pipeline to handle full data loads and change data (CDC) file from a Python process, store the data in AWS S3, and use Snowflake's native features — Snowpipe, Streams, Tasks, and Stored Procedures — to automatically load and update a Customer table with SCD Type 1 logic.

What is Slowly Changing Dimension(SCD) Type1?

Slowly Changing Dimensions are dimensions (like Customer, Product, Employee) whose attributes change slowly over time, not every day like transactions.

For example:

- A customer changes their email address.
- A product gets a new description.
- An employee updates their job title.

SCD Type 1 simply overwrites the old data with the new data.

- There is no history kept.
- Only the latest value is stored.
- So if something changes, the old value is lost forever.

Scenario

Source data (customer table)

Customer_id	Name	Email
1	John Doe	john@example.com

Target data (customer table)

Customer_id	Name	Email
1	John Doe	john@example.com

Scenario:

John updates his email to john.new@example.com.

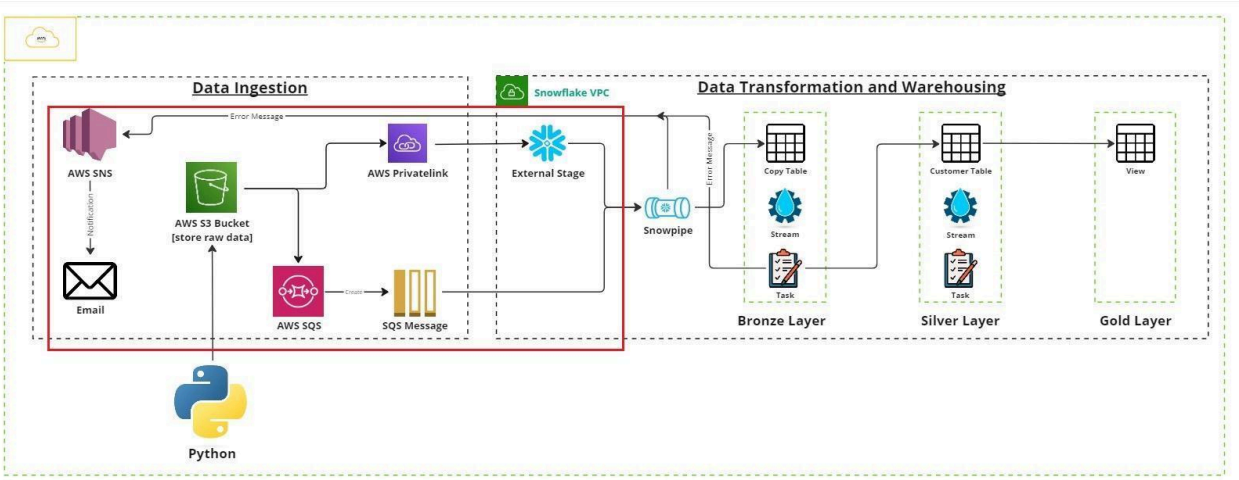
Source data (customer table)

Customer_id	Name	Email
1	John Doe	john.new@example.com

Target data (customer table)

Customer_id	Name	Email
1	John Doe	john.new@example.com

Architecture Diagram



How It Works

Data Upload

- A Python script (running in an Anaconda notebook) takes the downloaded full extracts and incremental change files from your local machine. These dataset is already shared as part of the project
- The script uploads these files to a designated S3 bucket.

Storage Integration

AWS S3 and Snowflake are connected via:

- Storage Integration: secure way for Snowflake to read files from S3.
- External Stage: defines the S3 location inside Snowflake.

Automated Data Load with Snowpipe

- Snowpipe continuously monitors the S3 stage.
- When new files arrive (full or CDC), Snowpipe automatically copies the data into a Customer Source table (CUSTOMER_SOURCE).

✅ Benefit: Fully automated, near real-time ingestion without manual loading.

Change Processing with Streams

- A Stream is defined on the CUSTOMER_SOURCE table.
- The Stream tracks new or changed rows that were loaded by Snowpipe.

✅ Benefit: Snowflake knows exactly which rows are new or updated, making incremental processing simple.

Automated Processing with Tasks & Stored Procedures

- A Task runs on a schedule (or continuously) and calls a Stored Procedure.
- The Stored Procedure:
  - Reads the change data from the Stream.
  - Merges it into the target CUSTOMER table using SCD Type 1 logic:
  - If the customer ID exists → update the changed attributes (e.g., overwrite address or email).
  - If the customer ID is new → insert it.
  - This is typically done using a MERGE statement.

✅ Benefit: The entire SCD1 flow is automated — no manual steps to merge data.

Benefits

- ☐ Automated full & incremental loads
- ☐ Zero manual file handling — everything runs on Snowflake side once data lands in S3
- ☐ No duplicate work — Stream ensures only new/changed rows are processed
- ☐ SCD1 logic guarantees the CUSTOMER table always has the latest values — old data is overwritten as required